Modelling framework of an instance of the electric power system: functional description and implementation

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Abstract. Critical Infrastructures (CI) are complex and highly interdependent systems, networks, and assets that provide essential services in our daily life. This paper addresses the analysis of the Electric Power Systems (EPS), among the major representatives of critical infrastructures, with focus on the interdependencies that strongly characterize these systems. First, the evaluation framework set up in the ongoing European project CRUTIAL is discussed. Then, we present the concrete EPS system under analysis, the functional description of the required models as well as their implementation using Stochastic Activity Networks formalism.

1 Introduction

Critical Infrastructures (CI) are complex and highly interdependent systems, networks, and assets that provide essential services in our daily life. They span a number of key sectors, including energy, finance, authorities, hazardous materials, telecommunications, information technology, supply services and many others. With our increasing dependence upon such critical infrastructures, an unavoidable expansion in complexity is observed since these sectors are continuously called to provide new services and products to a growing population. Moreover, the framework conditions under which these infrastructures operate are continuously evolving. In fact, while in the past they were used to provide services mostly in isolation, with very limited interconnections with each other, so they could only be impaired locally, nowadays several infrastructures cooperate in the provision of services. This implies that strong networking within and between sectors mainly trough information technology means is in place, whose dimension may also go beyond the national borders. The future trend is even more alarming since, among the others: i) disturbance phenomena will more and more affect large portions of critical infrastructures, ii) the variety and sophistication of threats is increasing, iii) incidents abroad are increasingly becoming a problem for the availability/security of the services provided by internal critical infrastructures.

Critical infrastructure protection is therefore a priority for most of the countries, and several initiatives are in place to identify open issues and research viable solutions in this highly challenging area, especially to identify vulnerabilities and devise survivability enhancements on critical areas. Identifying interdependencies and interoperabilities between systems is of primarily importance to understand what assets are the most critical ones. Following this analysis, steps can be taken to mitigate the identified vulnerabilities.

This paper addresses the evaluation of CI focusing on the Electric Power Systems (EPS), focusing in particular on the interdependencies between the information control system and the controlled electrical infrastructure. The offered contribution is twofold. First, the evaluation framework set up in the ongoing European project CRUTIAL is discussed, as a concrete research direction tailored to the electric power systems (Section 2). Then, we present the concrete EPS system under analysis, the functional description of the required models as well as their implementation using Stochastic Activity Networks formalism (Section 3). Final conclusions are drawn in Section 4.

2 A modeling framework for the quantitative EPS analysis under development in the CRUTIAL project

The CRUTIAL project [1] addresses new networked ICT systems for the management of the electric power grid, in which artefacts controlling the physical process of electricity transportation need to be connected with information infrastructures, through corporate networks (intra-nets), which are in turn connected to the Internet. One of the main objectives of CRUTIAL is the definition of modelling approaches for understanding and mastering the various interdependencies between the information control system and the controlled electrical infrastructure, thus providing both qualitative and quantitative support for the identification, analysis and evaluation of the identified critical scenarios.

The goal of this section is to provide an overview of the quantitative modeling framework that the authors of this paper are currently developing inside CRUTIAL. An overview on the qualitative modeling approach can be found in [2, 3]. The body of the modeling framework has been already introduced in [4], and a part of it is outlined here to provide a complete context overview and for better understanding the new developments presented in this paper. For this purpose, in Sections 2.1 and 2.2 we focus on the identification of the main logical components of the two infrastructures composing the Electric Power System (EPS): the Electric Infrastructure (EI) and the information infrastructure (ITCS). The interdependencies between ITCS and EI are then discussed in Section 2.3, while Section 2.4 summarizes the main features characterizing the modeling framework.

In [4], the authors also discussed the feasibility of the proposed framework using Möbius [5], a powerful multi-formalism/multi-solution tool, and presented the implementation of a few basic modeling mechanisms adopting the Stochastic Activity Network (SAN) formalism [6], which is a generalization of the Stochastic Petri Nets formalism. The goal was not to provide a complete model representing a concrete instance of an EPS system, but just to show how some basic frameworks characteristics can be actually obtained.

The last part of this paper starts attacking this pending aspect, and in particular Section 3 presents the concrete EPS system under analysis, the corresponding overall model, and it provides both a high-level functional description of the system's aspects captured by each submodel, and their concrete implementation using Stochastic Activity Networks formalism.

2.1 Logical scheme of EI

EI represents the electric infrastructure necessary to produce and to transport the electric power towards the final users. It can be logically structured in different components: the transmission grid (TG, operating in very high voltage levels), the distribution grid (DG, operating in medium/low voltage levels), the huge, medium and low voltage generation plants, and the huge, medium and low voltage loads.

The main elements that constitute the power grid are generators, loads, substations and power lines. One or more *generators* can be situated inside the power plants. The energy produced by the generators is then adapted by transformers, to be conveyed with minimal dispersion, to the different types of end users (*loads*), through different power grids. The *power lines* are components that physically connect the substations with the power plants and the final users, and the *substations* are structured components in which the electric power is transformed and split over several lines. In the substations there are transformers and several kinds of connection components (bus-bars, protections and breakers).

From a topological point of view, TG and DG can be considered like a graph: the nodes represent generators, substations, and loads, while the arcs represent the power lines.

V (voltage), Fr (frequency), I (current flow), A (angle), P (power) and Q (reactive power) are some of the main physical parameters associated to the electric equipments constituting EI (generators, substations, power lines and loads), and their specific values are of primary importance in determining the current status of the overall EI. In fact, they affect the behavior of the electric equipments they are referred to (e.g., in terms of availability and reliability of the electric equipment), thus also influencing the evolution of the overall power grid.

Figure 1(a) depicts the high-level logical scheme corresponding to a typical physical scheme of a substation with the connected power lines. The main electric equipments (bus-bars BB, protections PR, breakers BR, power lines PL) have been grouped following an approach which has the advantage to simplify the logical representation. The component N_S represents the parts common to all

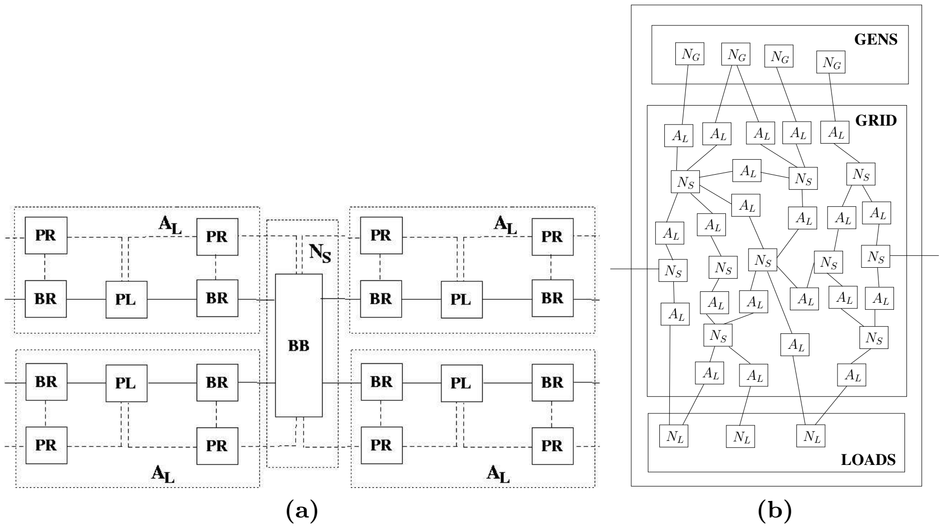

Fig. 1. Logical scheme (a) for substation and connected power lines, and (b) for a dummy transmission grid

substations (i.e., the bus-bar), while breakers and protections, which are physically part of a substation, are now included in the scheme of the new logical component A_L. In Figure 1(b) the corresponding high-level logical scheme for a dummy transmission grid is presented, where the components N_G and N_L represent a generation plant and a load, respectively.

2.2 Logical scheme of ITCS

$ITCS$ (Information-Technology based Control System) implements the control system based on information technology and its main purposes are: i) reducing out of service time of generators, power lines and substations (availability); ii) enhance quality of service (through frequency and voltage regulation); iii) optimizing generators and substations management. To these aims, ITCS performs the following activities: a) remote control of the electric infrastructure (it receives data and sends commands); b) co-ordination of the maintenance (it plans the reconfiguration actions that can affect generators, substations, loads and lines); c) collection of the system statistics. Among the several logical components composing ITCS, we focus the attention on the tele-operation systems for the distribution grid (named $DTOS$) and for the transmission grid (named $TTOS$), since a failure of these logical components can affect a large portion of the grid, also leading to black-out phenomena.

In Figure 2 we depict a possible logical structure of $TTOS$ and $DTOS$. The components LCS (Local Control System), RTS (Regional Tele-control System) and NTS (National Tele-control System) of $TTOS$, and the components LCC (Local Control Center), and ATC (Area Tele-control Center) of $DTOS$ differ for their criticality and for the locality of their decisions.

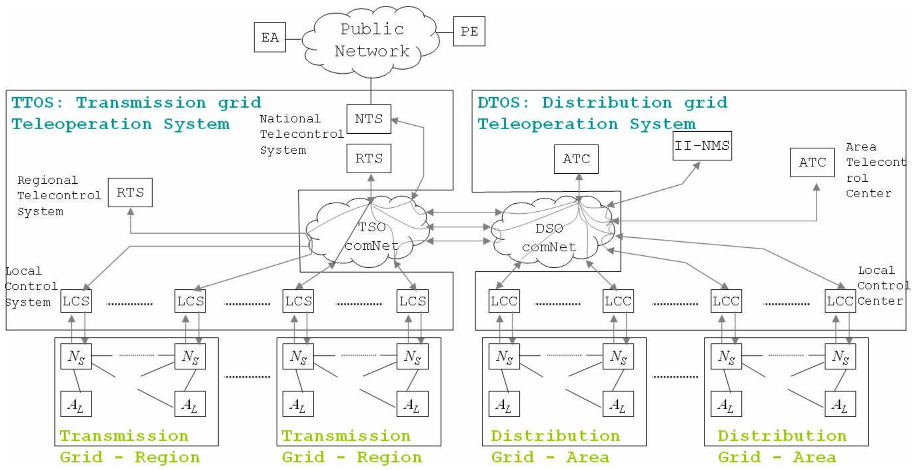

Fig. 2. Logical scheme of $TTOS$ and $DTOS$

Different actors (like Power Exchange PE, Energy Authority EA, Network Management System NMS) are involved in the electric system management and there can be a necessity to exchange grid status information and control data over public or private networks (e.g., $TSOcomNet$ and $DSOcomNet$). The transmission and distribution grids are divided in homogeneous regions and areas, respectively. LCS and LCC guarantee the correct operation of substation equipment and reconfigure the substation in case of breakdown of some apparatus. They include the acquisition and control equipment (sensors and actuators). RTS and ATC monitor their region and area, respectively, in order to diagnose faults on the power lines. In case of breakdowns, they choose the more suitable corrective actions to restore the functionality of the grid. Since RTS and ATC are not directly connected to the substations, the corrective actions to adopt are communicated to the LCS or ATC of reference. NTS has the main function of supervising the entire grid and handling the planning of medium and long term operations. NTS also assists RTS (and ATC) to localize breakdowns on the power lines situated between two regions (two areas). LCS and LCC, such as RTS and ATC, cooperate to decide operation of load shedding.

2.3 Interdependencies

An interdependency is a bidirectional relationship between two infrastructures through which the state of each infrastructure influences or is correlated to the state of the other. Among the possible types of interdependencies, here the focus is on the cyber and physical interdependencies. EI requires information transmitted and delivered by ITCS, for example when ITCS triggers a grid reconfiguration for economic optimization; therefore the state of EI depends on the outputs of ITCS (cyber interdependency). Viceversa, the state of ITCS could be affected by failures in EI (e.g. in case of blackout that leads to a failure or service degradation of the information infrastructure), thus revealing a physical interdependency.

Failures in ITCS impact on the state of EI, i.e. on the topology T and on the values of V, Fr, I, A, P and Q, depending on the logical components affected by the failures, and obviously by the type of the failures (cyber interdependency). For example, consequences of a failure of the component LCS associated to a component N_S, N_G or N_L (see also Figure 2) can be:

- Omission failure of LCS, fail silent LCS. No (reconfiguration) actions are performed on N_S or A_L.
- Time failure of LCS. The above (reconfiguration) actions on N_S or A_L are performed after a certain delay (or before the instant of time they are required).
- Value failure of LCS. Incorrect closing or opening of the power lines A_L directly connected to the failed component is performed.

The failure of the component RTS (or NTS) corresponds to an erroneous (request of) reconfiguration of the state of EI (including an unneeded reconfiguration) affecting one or more components of the controlled region. The effect of the failure of RTS (or NTS) on a component N is the same as the failure of the component LCS associated to the component N. In the case of Byzantine failure these effects can be different for each component N. In general, the failure of the components LCS, RTS and NTS may depend on the failures of the components connected to them through a network.

On the other direction (physical interdependency), failures of the EI infrastructure impact on (parts of) the ITCS system by lessening its functionalities (till complete failure in the extreme case the failure is a total blackout of the power grid).

2.4 Major characteristics of the EPS Modelling Framework

To represent and model the behavior of EI and ITCS and their interactions, the modelling and evaluation framework should possess a number of features encompassing the following aspects : i) modelling power, i.e., the basic modeling mechanisms required to build the EPS model; ii) modelling efficiency , i.e. the advanced modeling mechanisms required to build the EPS model more efficiently; and iii) solution power, i.e., the ability to provide efficient solution methods adequate for the EPS modelling complexity and for the assessment of the specific measures of interest. With specific reference to the structural and behavioral aspects of EPS systems, major requirements on a suitable modelling framework include:

1. The system has a natural hierarchical structure, as shown in the examples of logical schemes of Figure 2. Therefore, the modeling framework should support hierarchical composition of different sub-models. The model for the overall EPS could be facilitated considering replication of (anonymous and not anonymous[3]) sub-models, and the replicated and composed models should share part of the state (common state).
2. The state of EI is completely described through the physical parameters associated to each electric equipment (V, Fr, I, A, P and Q) and through the topology (T) of the grid: the first set of parameters defines the current status of each EI component, while the topology defines how such components are connected together to form the overall EI. Therefore, it is crucial that the modeling framework should support the representation of a hybrid-state composed by a discrete part (the topology) and a continuous one (the electric parameters).
3. The time to failures of the components N_S, N_G, N_L and A_L depends also on the value of the electric parameters associated to the components. This means that the framework should support time and probability distributions, and conditions enabling the time consuming events (e.g., for the activation of a local protection) that can depend both on the discrete and on the continuous state.
4. We need to consider the reconfiguration actions triggered by the ITCS components, e.g., by LCS and RTS. Moreover, the automatic evolution (autoevolution) of the electric parameters in case of instability events, e.g. in correspondence of a power line failure, should also be considered. Therefore, the framework should support the call to the functions implementing the reconfiguration algorithms, as well as the autoevolution algorithm.
5. To manage complexity at solution level, ability to perform separate evaluation of different sub-models and combination of the obtained results should be supported.
6. Risk analysis of EPS based on a stochastic approach requires the definition of measures of performability, which is a unified measure proposed to deal simultaneously with performance and dependability. To this purpose, a reward structure should be set-up by associating proper costs/benefits to generators/loads and interruption of service supply.

3 The analyzed EPS instance

In [4] the authors demonstrated the feasibility of the depicted modeling framework showing how its major characteristics (some of those detailed in Section 2.4) can be concretely implemented using a specific modeling formalism (SAN) and a specific modeling and solution tool (Möbius). Here we perform a further step: with reference to the concrete instance of the EPS that is currently under study, we identify the corresponding basic models and we describe their behavior from a functional point of view and their actual implementation. For the sake

[3] Not anonymous replicas can be identified by an index.

of simplicity, the proposed instance is limited to a homogeneous region of the transmission grid of EPS. Thus, the Local Control System (LCS) and the Regional Tele-control System (RTS) are only considered for ITCS. The considered region is composed by n_A power lines and n_N nodes, divided in n_G generators, n_L loads and n_S substations, with $n_N = n_G + n_L + n_S$. The topology of the network is described by the $n_A \times n_N$ *adjacency matrix* $\boldsymbol{A} = [a_{li}]$, where:

$$a_{li} = \begin{cases} 1 & \text{if line } l \text{ exits node } i, \\ -1 & \text{if line } l \text{ enters node } i, \\ 0 & \text{otherwise.} \end{cases}$$

The power associated to each node i is P_i, which is positive for the generators, negative for the loads and zero for the substations. The maximum power that a generator i can supply is P_i^{max} and the maximum power flow that a transmission line l can carry is F_l^{max}. A line is overloaded if the power flow exceeds F_l^{max}. The susceptance of each line l is b_l. Some simplifying assumptions have been made to represent the power flow through the transmission grid, following the same approach used in [7–10]. Therefore, the state and the evolution of the transmission grid are described by the active power flow $\mathbf{F}$ on the lines and the active power $\mathbf{P}$ of the nodes (generators, loads or substations) at the steady-state, i.e. when they have reached an equilibrium condition. $\mathbf{P}$ and $\mathbf{F}$ satisfy the linear equations for a direct current (`DC`) load flow approximation of the `AC` system:

$$\mathbf{P} = \boldsymbol{B} \cdot \boldsymbol{\Theta} \tag{1}$$

$$\mathbf{F} = \boldsymbol{D} \cdot \boldsymbol{A} \cdot \boldsymbol{\Theta} \tag{2}$$

with

$$\sum_{i=0,\cdots,n_N-1} P_i = 0, \tag{3}$$

where:

- $\boldsymbol{B} = \boldsymbol{A^T} \cdot \boldsymbol{D} \cdot \boldsymbol{A}$ is the $n_N \times n_N$ *susceptance matrix*, and $\boldsymbol{A^T}$ is the transpose of $\boldsymbol{A}$,
- $\boldsymbol{D} = diag(b_0, b_1, \ldots, b_{n_A-1})$ is the $n_A \times n_A$ diagonal matrix with the l-th diagonal entry representing the susceptance b_l,
- $\boldsymbol{\Theta} = (\theta_0, \theta_1, \ldots, \theta_{n_N-1})^T$, is the *node voltage angle vector*,
- $\mathbf{P} = (P_0, P_1, \ldots, P_{n_N-1})^T$ is the *node power vector*,
- $\mathbf{F} = (F_0, F_1, \ldots, F_{n_A-1})^T$ is the *line power flow vector.*

The initial setting of the distribution of the power produced by each generator is obtained by distributing the overall power demand to each generator proportionally to the maximum power that the generator can produce.

The operations performed by `ITCS` on `EI`, to control its correct functioning and activate proper reconfiguration in case of failure of, or integration of repaired/new, `EI` components are not considered in detail but they are abstracted at two levels, on the basis of the locality of the `EI` state considered by `ITCS` to decide on proper reactions to disruptions [10]. Each level is characterized by an activation condition (that specifies the events that enable the ITCS reaction), a reaction delay (representing the overall computation and application time needed by ITCS to apply a reconfiguration) and a reconfiguration strategy ($\mathcal{RS}$), based on generation re-dispatch and/or load shedding. The reconfiguration strategy $\mathcal{RS}$ defines how the configuration of EI changes when ITCS reacts to a failure. For each level, a different reconfiguration function is considered:

– $\mathcal{RS}_1()$, to represent the effect on the complete transmission grid of the reactions of ITCS to an event that has compromised the electrical equilibrium[4] of EI when only the state local to the affected EI components is considered. Given the limited information necessary to issue its output, $\mathcal{RS}_1()$ is deemed to be local and fast in providing its reaction. $\mathcal{RS}_1()$ is performed by LCS (or LCC) components, and it can be triggered by RTS (as actuation of a global reconfigurations) or can be directly triggered by LCS when it locally detects a non (electrical) equilibrium.
– $\mathcal{RS}_2()$, to represent the effect on the complete transmission grid of the reactions of ITCS to an event that has compromised the electrical equilibrium in EI when the state global to all the EI system under the control of ITCS is considered. Therefore, differently from $\mathcal{RS}_1()$, $\mathcal{RS}_2()$ is deemed to be global and slower in providing its reaction. $\mathcal{RS}_2()$ is performed by RTS.

The activation condition, the reaction delay and the definition of the functions $\mathcal{RS}_1()$ and $\mathcal{RS}_2()$ depend on the policies and algorithms adopted by $TTOS$.

An autoevolution function $\mathcal{AS}()$ is also considered to represent automatic evolution of EI when an event modifying the grid topology occurs. In this case, EI tries to find a new electrical equilibrium for the new grid topology, by changing the values of the power flow through the lines but leaving the generated and consumed power unchanged (only redirection of current flows). The new values for the power flow through the lines depend on the power injected on the nodes of the grid, on the electrical characteristics of the power lines (e.g., the susceptance) and on the topology of the grid.

The autoevolution is triggered each time an event that modifies the grid topology occurs (typically, a disruption of an EI component or the integration of a repaired/new EI component in the electric grid). The new equilibrium is reached instantaneously (if any) and no local LCS actions are performed. Otherwise, LCS operations are triggered to generate the new values for **P** and **F**, modelled through evaluating the reconfiguration strategy function $\mathcal{RS}_1()$. In both cases, (with or without success of the autoevolution) RTS is triggered to generate the new values for **P** and **F**, based on a blobal view of the system, by evaluating $\mathcal{RS}_2()$. Moreover, the autoevolution $\mathcal{AS}()$ and the reconfiguration strategies can be also triggered after a transient failure or a repair of LCS or RTS, because could be needed to apply an update of the configuration of EI.

[4] Events that impact on the electrical equilibrium are typically an EI component's failure or the insertion of a new/repaired EI component; for simplicity, in the following we will mainly refer to failures.

The functions $\mathcal{AS}()$, $\mathcal{RS}_1()$ and $\mathcal{RS}_2()$ are based on the state of EI at the time immediately before the occurrence of the failure. The reconfiguration strategy $\mathcal{RS}_1()$ is applied immediately, $\mathcal{RS}_2()$ is applied after a time needed to RTS to evaluate it. When new events occur changing the status of EI during the evaluation of $\mathcal{RS}_2()$, then the evaluation of $\mathcal{RS}_2()$ is restarted based on the new topology generated by such events.

In the considered instance, the output values of $\mathcal{AS}()$ for active power flow $\mathbf{F}$ on the power lines are derived by solving the linear power flow equation system (1), (2) and (3) for fixed values of $\mathbf{P}$. The output values of $\mathcal{RS}_1()$ and $\mathcal{RS}_2()$ for $\mathbf{P}$ and $\mathbf{F}$ are derived considering that for a given power demand, the power flow equations do not have a unique solution. The adopted definition for the function $\mathcal{RS}_1()$ is given by the solution (values for $\mathbf{P}$ and $\mathbf{F}$) of the power flow equations (1), (2) and (3) while minimizing a simple cost function, indicating the cost incurred in having loads not satisfied and having the generators producing more power. The output values of $\mathcal{RS}_2()$ for $\mathbf{P}$ and $\mathbf{F}$ are derived by solving an optimization problem to minimize the change in generation or load shedding with respect to the initial configuration at time 0, considering more sophisticated system constraints, as described in [10]. The power flow equations and the cost function are defined such that the following requirements are satisfied:

- Dispatch of generator is preferred to shedding of loads.
- Variation of the power produced by not failed generators is proportionally to the maximum power that they can produce.
- When a generator or a load are failed the associated power is forced to 0.
- When a line is disconnected (at least one breaker is open) the power flow is forced to 0.
- When `LCS` associated to a generator is affected by omission failure no dispatch is performed. The produced power changes to 0 only if the protection fires or if the protection does not fire and the generator fails.
- When `LCS` associated to a load is affected by an omission failure no shedding is performed.
- When `LCS` associated to generators and loads are failed the variation of the power P on the generators is forced, to represent the stress of the generator.
- The value of $\mathbf{P}$ is not fixed for a generator or a load associated to omission failed `LCS`, because the variation of power $\mathbf{P}$ have to be evaluated if the if the generator or the load are disconnected.
- The power flows $\mathbf{F}$ on the lines are not limited for $\mathcal{RS}_1()$.
- The absolute value of the power flow $\mathbf{F}$ on the line l is limited by F_l^{max} for $\mathcal{RS}_2()$.
- The power $\mathbf{P}$ on a generator is limited by the maximum power that the generator can produce.
- The power $\mathbf{P}$ on a load is limited by the demand.

As far as the measures of interest are concerned, we are interested in performability measures on the basis of a reward structure where costs and rewards are considered with respect to the point of the view of the power producers and distributers. Among them: i) the expected reward at a given instant of time or in a given time interval; ii) the expected percentages of blackouts at a given instant of time or in a given time interval; iii) the expected numbers of EI components affected by a failure at a given instant of time or in a given time interval.

3.1 The SAN formalism and the tool Möbius

The SAN formalism [6] is a stochastic extension to Petri nets based on four primitive objects: places, activities, input gates, and output gates. Places and activities represent the state and the actions of the modeled system, respectively. In particular, places represent short data types and special places, called extended place, allow the representation of the primitive data types typical of the programming language C++, like int, float, double, including structures and arrays of primitive data types. Input gates control the enabling of activities and define the marking changes that will occur when an activity completes (fires). Output gates define the marking changes of the state of the system when an activity completes. The attributes of the SAN primitives are defined by using sequences of C++ statements.

The software tool Möbius [5] supports multiple high-level modeling formalisms (including the SAN) and multiple solution techniques. It also allows the construction of composed models from previously defined models, supporting a hierarchical approach to modeling based on state-sharing. In this approach the submodels are linked together through sharing of state variables of each submodel. A state variable represents some portion of the state of a model, and is represented differently by different formalisms. In a SAN model (and in general for various stochastic extensions to Petri nets), a place represents a state variable. Thus, it is possible to compose two SAN models by holding a particular place (state variable) in common. Möbius features the Replicate/Join composed model formalism. A Join is a state-sharing composition node used to compose two or more submodels. A Join node may have other Joins, Replicates, or other submodels defined as its children. A Replicate is a special case of the Join node used to construct a model consisting of a number of identical copies of a submodel. State variables (with the same name in each of the instances) can be held in common among all replicated instances of the submodel. In the framework of Möbius, the most basic model is called an atomic model.

3.2 The sub-models composing the overall EPS model

In modelling the considered EPS, we followed a modular and compositional approach. The following atomic models (the leaves in Figure 3) have identified as building blocks to generate the overall EPS model.

- PL_SAN, which represents the generic power line with the connected transformers.
- PR1_SAN and PR2_SAN, which represent the generic protections and the breakers connected to the two extremities of the power line (see Figure 1(a)).

– N_SAN and LCS_SAN, which represent, respectively, a node of the grid (a generator, a load or a substation) and the associated Local Control System LCS (see Figure 1(b)).
– AUTOEV_SAN and RS_SAN, which represent, respectively, the automatic evolution of EI when an event modifying its state occurs, and the local reconfiguration strategy applied by LCS (function $\mathcal{RS}_1()$). Moreover in RS_SAN is defined the initialization code of the overall model at time 0.
– RTS_SAN and COMNET_SAN, which represent, respectively the Regional Telecontrol System RTS, where the regional reconfiguration strategy $\mathcal{RS}_2()$ is modeled, and the public or private networks (e.g., $TSOcomNet$ and $DSOcomNet$ of Figure 2).

In Figure 3, it is shown how the atomic models are composed and replicated to obtain the composed model representing the EPS region.

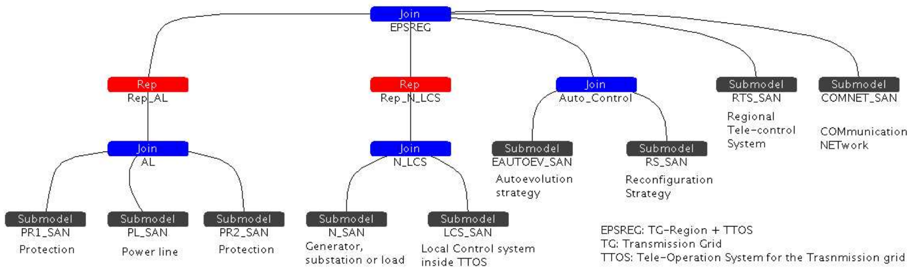

Fig. 3. Composed model for an EPS region

The submodel AL represents a power line with the associated protections and it corresponds to logical components A_L of Figure 1(b). This model is a *Join* node obtained by composing the SAN atomic models PR1_SAN, PL_SAN and PR2_SAN. The *Replicate* node Rep_AL is the submodel obtained by replicating n_A times the submodel AL to obtain all the necessary non anonymous components A_L of the grid. The model N_LCS is a *Join* node obtained by composing the atomic models N_SAN and LCS_SAN. The *Replicate* node Rep_N_LCS is the submodel obtained by replicating n_N times the model N_LCS to obtain all the necessary non anonymous N_G, N_S and N_L components of the grid, with the associated LCS. The model Auto_Control is a *Join* node obtained by composing the atomic models AUTOEV_SAN and RS_SAN, so it represents both the autoevolution function $\mathcal{AS}()$ and the reconfiguration strategy $\mathcal{RS}_1()$ locally applied by the LCS components. The overall EPSREG model is a *Join* node finally obtained through composition of the different models and it represents the EPS instance under study.

The different (atomic or non atomic) submodels interact each other sharing some common (extended or not extended) places that represent the parameters or part of the states of the EPS, like:

- The topology of the grid, modeled by two extended places `NodeOfArc` and `ArcOfNode`. `NodeOfArc` is an array of n_A struct type elements (i, j), one for each line l of the grid. The short fields i and j represent the indexes of the nodes associated to a line l oriented from i to j. `ArcOfNode` is an array of $n_N \times (n_N + 1)$ short elements. `ArcOfNode` associates to each node i of the grid, an array `ArcOfNode`–$>Index(i)$ of $(n_N + 1)$ elements representing the list of the indexes of the lines of the grid associated to i. The first element `ArcOfNode`–$>Index(i)$–$>Index(0)$ is the number of lines, the other elements, from `ArcOfNode`–$>Index(i)$–$>Index(1)$ to `ArcOfNode`–$>Index(i)$–$>Index($ `ArcOfNode`–$>Index(i)$–$>Index(0))$, represent the indexes of the lines.
- The susceptance of the lines, modeled by the extended places $b0$ and $B0$ for the values at time 0 and by the extended places b and B for the current values at time $t > 0$. b and $b0$ are arrays of n_A double type elements. Each element b–$>Index(l)$ represents the susceptance b_l of the line l. B and $B0$ are arrays of $n_N \times n_N$ double type elements, representing the susceptance marix.
- The power of the nodes of the grid $\mathbf{P}$, modeled by two extended places `P0` and `P`, respectively for the values at time 0 and at the current time $t > 0$. `P0` and `P` are arrays of n_N double type elements. Each element `P`–$>Index(i)$ or `P0`–$>Index(i)$ of the arrays represents the power associated to each node i of the grid.
- The power flow through the lines of the grid $\mathbf{F}$, modeled by the extended place `F`, that is an array of n_A double type elements. Each element `F`–$>Index(l)$ of the array represents the power associated to the line l of the grid.
- The results for $\mathbf{P}$ and $\mathbf{F}$ of the evaluation of $\mathcal{RS}_2()$, modeled by the extended places `P_SR2` and `F_SR2`, respectively. These results are derived in the model `RS_SAN` and are "passed" to the model `RTS_SAN`, where they are applied to the grid.
- The temperature of the lines and the instant of time when the last update of the temperature is occurred, modeled by the extended place `Tl`, that is an array of n_A struct type elements $(doubleTemperaturel, doubletimel)$.
- The open lines because the breaker is open, modeled by the extended place `Open`, that is an array of n_A short type elements. Each element `Open`–$>Index(l)$ of the array is equal to 2, 1 or 0 if both protections of the line l are open, only one protection is open or no protection is open, respectively.
- The number of open lines, modeled by the place $nOpen$.
- The lists of the failed nodes or lines, modeled by the extended places `EOutageNlist` and `EOutageAlist`, respectively. They are arrays of $n_A + 1$ and $n_N + 1$ short type elements, respectively. The first element of these arrays is the number of failed elements, the other elements represent the indexes of the failed components.
- The variation of the power through a node (generator or load) after a failure, modeled by the extended place `deltaP`, that is an array of n_N double type elements.
- The trigggering of the failure of a generator and of the firing of the associated protection, caused probabilistically by the variation of the power required to

a generator after a failure. This status for all the generators is modeled by the extended place deltaPFail, that is an array of n_G short type elements. When deltaPFail–>$Index(i)$ is set to 1 the generator i will fail in a deterministic time t_{DF} if the associated protection will not fire in a deterministic time less than t_{DF}. The protection could not fire if it is failed.

- The failed LCS, modeled by the extended place LCSfailure, that is an array of n_N struct type elements $(shortomission, shortvalue)$. An omission or value failure of the instance of LCS associated to the node i is represented by setting to 1 LCSfailure–>$Index(i)$–>$omission$ or LCSfailure–>$Index(i)$ –>$value$, respectively.
- The list of LCS failed during the evaluation of $\mathcal{RS}_2()$, modeled by the extended place LCSnewFailure. It is an array of $n_N + 1$ short type elements. The first element of this array is the number of failed elements, the other elements represent the indexes of the failed components. After RTS has applied the new reconfiguration $\mathcal{RS}_2()$, if new failures of LCS are occurred during the evaluation of $\mathcal{RS}_2()$, the autoevolution or $\mathcal{RS}_1()$ are triggered, because some generator or load could not change its configuration for **P**.
- The status of the occurrence of a propagation of a failure or a lightning, modeled by 0 token in the place PropagLock, that is set to 1 at time 0.
- The failure rates of the nodes and of the power lines, modeled by the extended places NEFailureRates and AEFailureRates, respectively. They are arrays of n_N and n_A struct type elements, respectively. Each struct type has different fields for the different types of failures considered for nodes and power lines.

These models populate our modelling framework as template models, which are used to represent a large variety of specific scenarios in the EPS sector. Theoretically, all the possible EPS configurations involving (a subset of) the addressed components are representable through proper combination of the proposed models (unless some aspects have been currently not yet captured. Exercising the developed framework on several different scenarios will be useful to reveal possible aspects not included and then proceed with a refinement).

In the following we provide some more details on the system's aspects captured by each atomic model. The focus is especially on failure events and on how they propagate among interacting components/subsystems.

The atomic model PL_SAN

PL_SAN, shown in Figure 4, represents the generic power line (and the connected transformers, if any). This SAN is replicated n_A times. Each instance of PL_SAN represents one of the n_A power lines of the considered system. The index of a power line is represented by the number of tokens in the local place ALindex, that is set when the immediate activity setupIndex completes by the otuput gate setIndex, which is defined as:

```
ALindex->Mark()=(nA-ALcount->Mark())-1;
```

The place Start is initialized with one token. The common place ALcount is shared with all the replicated instances of the submodel and with RS_SAN. The immediate activities setupIndex of the replicated instances are all enabled in the same marking at time 0, when the model RS_SAN set to n_A the number of token of the common place ALcount. Thus, the first activity setupIndex which completes set the ALindex of the associated instance to 0 (one token is removed by the input place ALcount and the code of setupIndex is executed), the next activity setupIndex which completes set the ALindex of the associated instance to 1, and so on.

PL_SAN models the power flow through the power line (with the extended place F), the instant of time and the temperature associated to the last change of the power flow (with the extended place Tl), the states where the power line works correctly (with 1 token in the local place EWorking) or is affected by a failure (with 1 token in the local place EOutage) and the list of the failed power lines (with the extended place EOutageAlist).

Moreover, the following main states or parts of states are modelled:

- A transient failure is occurred, modeled by a token in the local place ETempFailureLi (for a lightning) or ETempFailure (for the other failures).
- A permanent failure is occurred, modeled by a token in the local place EPermFailure.
- The power of the occurred lightning, modeled by the value of the extended place LightningF with type double.
- The power of the lightning to propagate to nodes i and j at the ends of the line, represented by the extended places LiFtoi Liftoj with type double, respectively.
- The power of the lightning propagated from the models PR1_SAN and PR2_SAN of protections i and j (or 1 and 2) to the line, represented by the extended places LiFiTol or LiFjTol with type double, respectively.
- The evolution (the state) of the propagation, modeled by the local place Propag (one token in this place if the propagation is occurring), the local place PropagateByClosed (one token in this place if the propagation of failure has been triggered by a closure of a protection), the place CountBranch and the extended place LiF.
- The lock of the propagation, modeled by the extended place PropagLock.
- The triggering of the propagation caused by the closure of a breaker, modeled by the extended place PropagateLine, that is an array of n_A short type elements. The propagation of the failure from a line l is triggered when PropagateLine−>$Index(l)$ is set to 1 (by an instance of the models PR1_SAN or PR2_SAN) and the place Propag is set to 0 (no other propagation is occurring).
- The triggering of the autoevolution and of the reconfiguration strategy $\mathcal{RS}_2()$, modeled by one token in the places autoev and RS2 at the end of a propagation of failure or lightning.

Four exponential activities represent the times to the failures of the power line (and of the connected transformers). Two cases associated to these activities represent the probabilities of permanent or transient failure. The considered failures are:

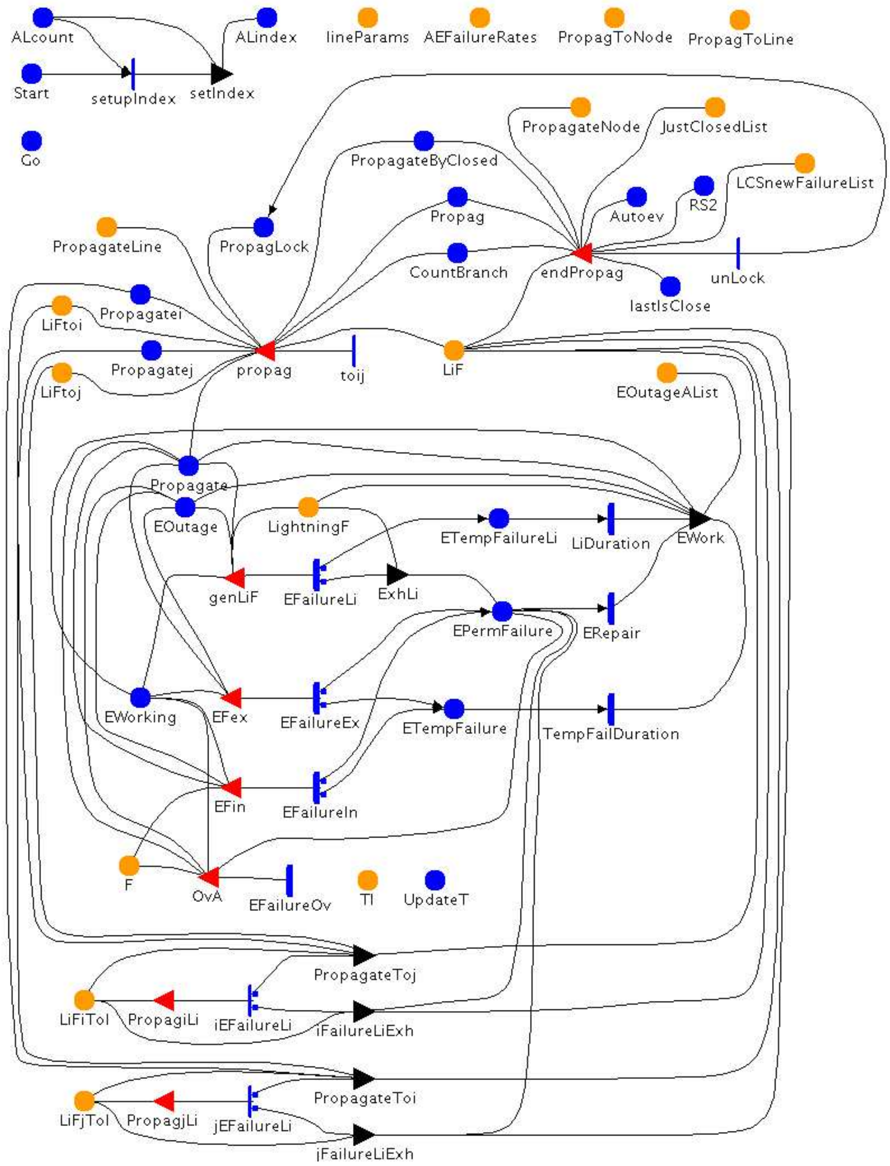

Fig. 4. The model PL_SAN for a generic power line

– External failure caused by a lightning, modeled by `EFailureLi`.
– External failure not caused by a lightning, modeled by `EFailureEx`.
– Permanent internal failure caused by an overload, modeled by `EFailureIn`.
– Internal failure not caused by an overload, modeled by `EFailureOv`.

The failure rates are based on the values of the extended place `AEFailureRates`. The rate of the internal failures are set with a function depending on the power flow of the line. In particular, the rate for the internal failure not caused by an overload is constant when the power flow is below the threshold for which an overload occurs and increases linearly when the power flow is greater than this threshold. The rate for the overload failure also depends on the temperature of the line, following the same approach proposed in [9] for modeling the heating.

When the activity representing the occurrence of the lightning completes, the input gate set the place `LightningF` with a value generated randomly from an uniform distribution.

The probability that a lightning produces a permanent failure (the break down of the line) depends on the power of the lightning. If the failure is transient (the line does not break down) and the breakers do not open, the lightning can propagate to the ending nodes i and j of the power line (e.g., half power of lightning moves toward node i and half power moves toward node j). When the lightning passes through a line or a node of the grid it reduces its power. The lines, the nodes or the breakers passed through by a lightning can fail (break down) depending on the power of the lightning and on the power flow through them. The propagation ends when the protections stop the propagation,or the lightning reaches a failed line or node, or when the lightning exhausts its power, i.e., `LiF`$->Index(l)->Mark() == 0$). If the lightning produces a permanent failure of a line or a node, it exhausts and does not propagate anymore.

When the lightning produces a permanent failure and for each other type of failure, a propagation of the failure is considered (the place `CountBranch` is set to 2). The failure propagation starts from the failed line (moving towards the nodes at the two ends) or node (moving towards the connected lines). For each fork of the propagation from a node towards a new line one token is added to the place `CountBranch`. A branch of the failure propagation stops when: i) a protection fires (opening a breaker), ii) when the propagation reaches a node already touched by the propagation process, or iii) the propagation reaches a failed line or node. In these cases a tooken is removed from the place `CountBranch`. The overall propagation ends when it cannot more propagate, i.e. when `CountBranch`$->Mark() == 0$.

Therefore, two types of *propagation* are considered when a failure occurs: the propagation of a lightning and the propagation of a failure in output toward the ending nodes i and j (represented by two instances of the models `N_SAN`), passing for the models `PR1_SAN` and `PR2_SAN`, respectively. The effect of the propagation of a lightning or a failure is to isolate the failed component from the rest of the grid. If the propagation reaches a generator or a load, because the protections did not fire, these components are considered failed. The propagation is instantaneous and when it occurs, the other propagations (triggered by failures of other components) cannot start until the current propagation ended. The propagations are represented by the enabling and completion of immediate activities and by places common to different submodels. In particular, when a failure occurs in the line, the immediate activity `toij` is enabled that locks the propagation (by subtracting one token to the place `PropagLock`) and adds one token to the places `Propagatei` and `Propagatej`. Moreover, in the case of lightning propagation also the extended places `LiFi` and `LiFj` are set to the half of the power of the lightning (half power of lightning moves to i node and half power moves to j node).

When the marking representing the end of the propagation is reached the immediate activity `unLock` is enabled that unlocks the propagation (by adding one token to the place `PropagLock`). Moreover, the autoevolution $\mathcal{AS}()$ and the reconfiguration strategy $\mathcal{RS}_2()$ are triggered, if the state of `EI` did change (see below the reconfiguration triggered by the closure of a breaker).

The duration of a transient failure is represented by the timed activities `LiDuration` (deterministic for the lightning) and `TempFailDuration` (exponential for the other failures). The repair time of a permanent failure is represented by the deterministic activity `ERepair`. When the transient failure expires or the permanent failed line is repaired the line is working, but the power flow through this line is zero until the protections close the breakers that isolate this line from the grid.

Two immediate activities `iEFailureLi` and `jEFailureLi` are enabled when a lightning propagates from the model `PR1_SAN` or `PR2_SAN`, respectively. Two cases on the activities represent the probability that the lightning causes a failure of the line (if it is not already failed). In the case of failure of the line, the lightning exhausts its power, otherwise the lightning moves to the model `PR2_SAN` (or to the model `PR1_SAN`) with a reduced power, represented by the extended places `LiFj` (or `LiFi`).

Finally, the propagation of the failure is triggered for each failed line every time that a protection closes a breaker (e.g., after a repair of the breaker). In this way, if a breaker closes when the effect of a failure is not yet exhausted, the breaker has to reopen. In this case at the end of the propagation the autoevolution $\mathcal{AS}()$ and the reconfiguration strategy $\mathcal{RS}_2()$ are not triggered, because the state of EI did not change. To trigger this propagation, the input gate `propag` of the immediate activity `toij` check the value corresponding to the line of the extended place `PropagateLine`, defined as an array of n_A short. This value is set in the models `PR1_SAN` and `PR2_SAN` when a breaker close. Moreover, the extended place `JustClosedList` represents the list of the breakers closed at the same time. This list can be updated during the propagation of failure triggered by the closure of the protections, if the closed protection must reopen. This list is used at the end of the propagation to verify if all the closed breaker have been reopen or not.

The atomic models PR1_SAN and PR2_SAN

`PR1_SAN`, shown in Figure 5, and `PR2_SAN` represent the states and the behavior of the generic protections and the breakers connected to the two extremes of the power line (see Figure 1(a)), and they are similar.

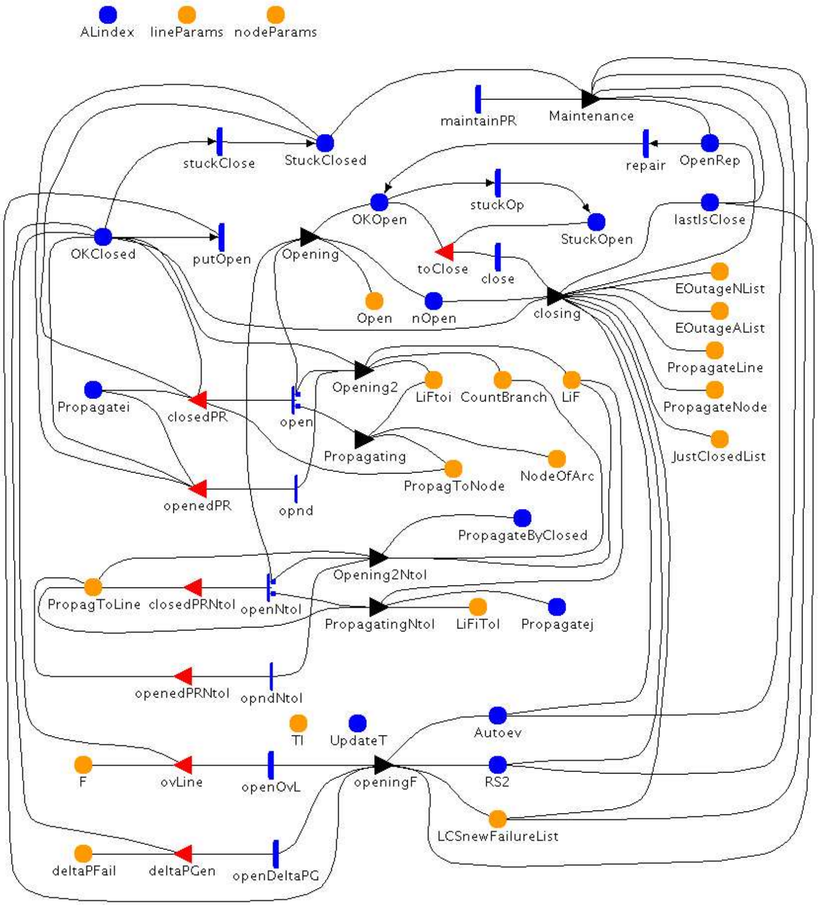

Fig. 5. The model `PR1_SAN` for a generic protection/breaker component associated to the end i of a power line

The following states of the breaker connecting a line and a node are modeled:

– Closed (when the line and the node are connected), modeled by 1 token in the place `OKClosed`.
– Stuck closed (when the line and the node are connected but, due to a failure, the breaker cannot open), modeled by 1 token in the place `StuckClosed`.
– Opened (when the line and the node are not connected and the power cannot flow), modeled by 1 token in the place `Open`.

- Stuck opened (when the line and the node are not connected but, due to a failure, the breaker cannot close), modeled by 1 token in the place `StuckOpen`.
- Open to repair (when the breaker is open during the repair), modeled by 1 token in the place `OpenRep`.
- The triggering of the protection caused by the variation of power of the generator connected to the protection after a failure, modeled by the extended place `deltaPFail`.

The times to the occurrence of a stuck open failure, stuck closed failure or to erroneously open the closed line depend on the power flowing through the line (if any), and are modeled by the exponential activities `stuckOp`, `stuckClose` and `putOpen`, respectively. The times to maintenance, to repair and to close an opened breaker are modeled by the deterministic activities `maintainPR`, `repair close`, respectively. To start the repair of a stuck closed breaker, the breaker is forced to open. The deterministic activity `openOvL` models the time to open the breaker when the line is overloaded. In this case, i.e., when the power flow through the line is greater than a threshold (depending on the line), after a deterministic time, the protection fires and the breaker opens, if it is not stuck closed. This time is based on the temperature of the line. The breaker should open some second before the temperature of the line becomes greater than a critical threshold. The time associated to the activity `openOvL` is restarted every time that the temperature of the line is updated (by using the reactivation memory mechanism), i.e. when the place `UpdateT` is set to 1 by the `RS_SAN` or `RTS` after a new reconfiguration has been applied. The deterministic activity `openDeltaPG` models the time to open the breaker when the connected generator can fails because a big variation of the demand of power. When, after a failure, the variation of power required from the generator connected to the protection is greater than a threshold or the associated LCS is affected by an omission failure, after a deterministic time the protection fires and the generator is disconnected from the grid. In these cases, when the breaker open, the autoevolution $\mathcal{AS}()$, the reconfiguration strategy $\mathcal{RS}_2()$ (the place `RS2` is set to 1) and, if needed, the reconfiguration strategy $\mathcal{RS}_1()$ (the place `autoev` is set to 1) are triggered and the list represented by the extended place `LCSnewFailureList` is reset to empty.

A breaker can be closed only if it is open and if it can be maintained closed, and it can be maintained closed if the failures that leaded to its opening have been removed, otherwise the breaker remains open. After the closure of a breaker, the propagations of failure are triggered for each failed line and node of the grid. When these propagations end, the breaker is reopened, if the effect of the failures on the protection is not exhausted. Otherwise, if there are not more failures that impact on the protection, the state of `EI` can change (for the closure of the breaker) and the autoevolution $\mathcal{AS}()$, the reconfiguration strategy $\mathcal{RS}_2()$ and, if needed, the local reconfiguration strategy $\mathcal{RS}_1()$ are triggered the list represented by the extended place `LCSnewFailureList` is reset to empty.

Propagation of a failure and of a lightning can move in input toward the model PR1_SAN from the connected line or node, i.e., from the models PL_SAN or N_SAN, respectively, and can cause the opening of the breaker, if it is not failed (stuck closed). This propagation is represented by the completion of the immediate activities open (in the case that the breaker is closed) and opnd (in the case that the breaker is already open). Depending on the power of the lightning, the propagation can also probabilistically cause the stuck closed failure of the breaker, if it is not already failed. This probabilistic choise is represented by the cases associated to the activity open. In the case of failure, the lightning continues the propagation in output toward the model N_SAN, if it come from PL_SAN (and only if the node has not been already touched), or viceversa.

To model the propagation from the models PR1_SAN or PR2_SAN to the model N_SAN and from N_SAN to the models PR1_SAN or PR2_SAN related to the output lines, the extended place PropagToLine defined as array of n_N struct type element is considered. This place is common to different submodels and represents for each node the occurrence of the propagation of a failure or a lightning.

The atomic model N_SAN

The SAN N_SAN, shown in Figure 6, represents the states and the behavior of the generic node of the grid $\mathtt{N_S}$ (generator, substation or load). This SAN is replicated n_N times. Each instance of N_SAN represents one of the generators, substations or loads of the considered system. The index of a generator, substation or load is represented by the number of tokens in the local place Nindex, following the same approach used for PL_SAN. N_SAN models the power of each node of the grid $\mathbf{P}$, the states where the logical component $\mathtt{N_S}$ (a node of the grid) works correctly or is affected by a permanent or transient failure and the list of the failed nodes of the grid. The considered failures are the same of PL_SAN, except for the failure caused by an overload.

Like for the power lines, two types of *propagation* are considered when a failure occurs: the propagation of a lightning and the propagation of a failure in output toward all the connected lines (passing for the protection models PR1_SAN or PR2_SAN, depending on the logical direction of the line). Moreover, the propagation of a failure and of a lightning can move in input toward the model N_SAN from a connected line (enabling the immediate activity propagNode), and the lightning propagation can probabilistically cause the failure of the node if it is not already failed (probabilistically modeled by the cases of the activity). In this case, the lightning exhausts its power, otherwise the lightning moves with a reduced power toward the models of all the connected lines (passing through the protection models PR1_SAN or PR2_SAN, depending on the logical direction of the line). The lightning propagates toward the output lines if it did not exhaust its power passing through the node. The lightning power propagated in output towards each line is proportional to the value of the susceptance of the output lines.

For the instance of the model representing a generator, when, after a failure, the variation of power required by the generator is greater than a threshold, or

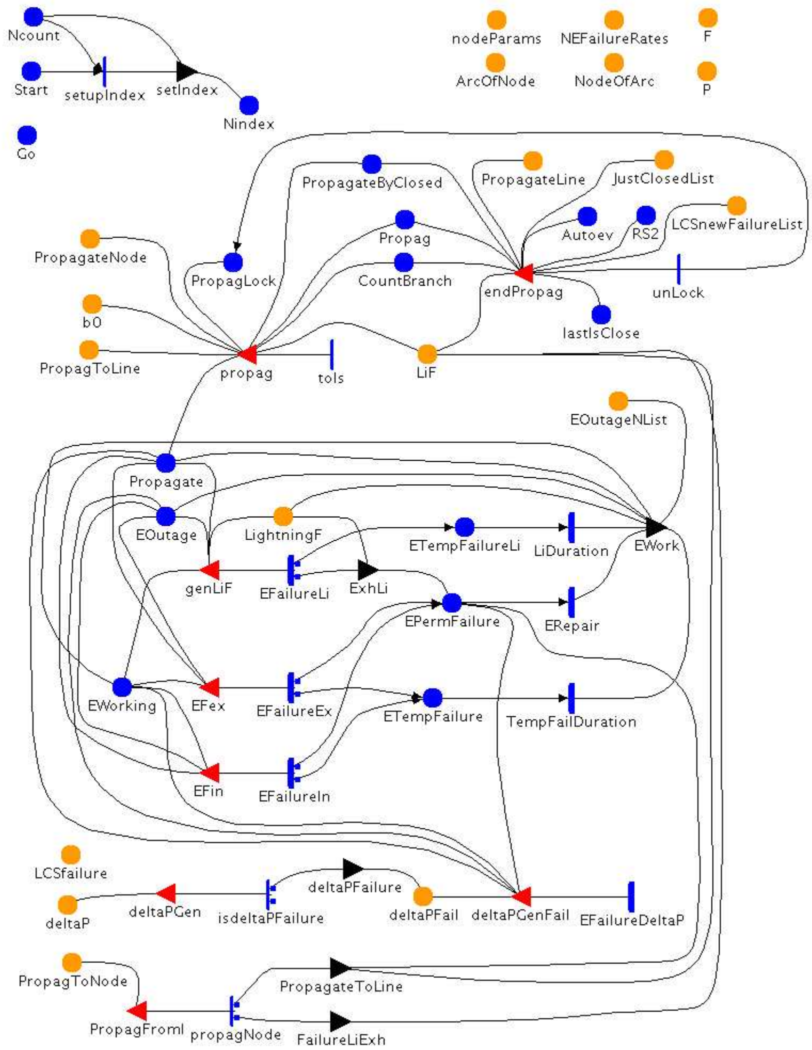

Fig. 6. The model N_SAN for a generic station

the associated LCS is affected by an omission failure, the protection is triggered to fire after a deterministic time and the generator fails after a deterministic time, if the protection did not fire. If the variation of power required by the generator is below a threshold, the protection is probabilistically triggered depending on the value of the variation. In this way, it is probabilistically modeled the probability that variations of the power required to a generator can cause the disconnection from the grid or the failure of the generator.

The model N_SAN can trigger the autoevolution $\mathcal{AS}()$, the reconfiguration strategy $\mathcal{RS}_2()$ and, if needed, the local reconfiguration strategy $\mathcal{RS}_1()$. In this case, the list represented by the extended place LCSnewFailureList is also reset to empty.

The atomic model LCS_SAN

LCS_SAN, shown in Figure 7, represents the states and the behavior of the Local Control System inside TTOS associated to a node. In particular, it models the states where LCS works correctly (1 token in the local place S_LCS) and the states where omission failure occurs (1 token in the local place omissionF, the only kind of failures considered at the moment for ITCS). Moreover, the list of the LCS that are failed during the evaluation of the reconfiguration strategy $\mathcal{RS}_2()$ is modeled by the extended place LCSnewFailureList. The information about the failed components is represented by the extended place LCSfailure.

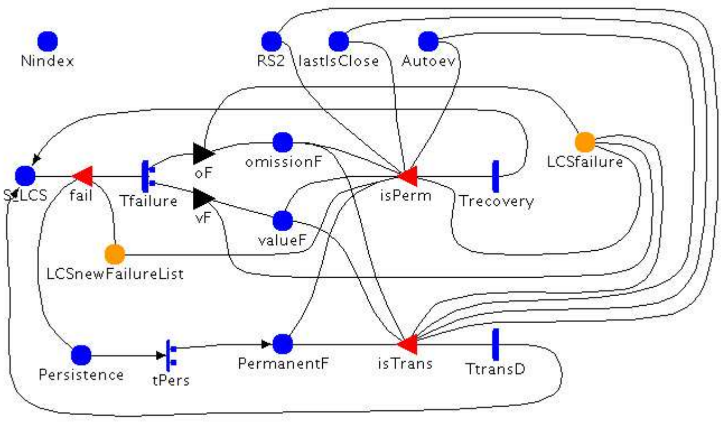

Fig. 7. The model LCS_SAN for a generic LCS component

The time to failure is represented by the exponential activity Tfailure. When LCS is affected by an omission failure, the generator or load controlled by it cannot be reconfigured by assigning proper power values. Failures can be transient (0 token in the local place PermanentF) or permanent (1 token in the local place PermanentF), depending probabilistically on the case selected when the activity `tPers` completes. The duration of the permanent failure is equal to the deterministic time to repair the component LCS and is modeled by the activity `Trecovery`. The transient failure exhausts after an exponential time represented by the activity `TtransD`.

The autoevolution $\mathcal{AS}()$ is triggered after a transient failure or a repair of LCS, because could be needed to apply an update of the configuration of EI. The detection of the LCS failure is immediate (time to detection is assumed 0). Thus when the evaluation of $\mathcal{RS}_2()$ starts, all the failed LCS components, for which P cannot change, are considered.

The atomic models AUTOEV_SAN and RS_SAN

`AUTOEV_SAN`, shown in Figure 8, and `RS_SAN`, shown in Figure 9, do not represent the behavior of a single component of EPS. They model, in terms of generation, re-dispatch and load shedding, the effect on the complete transmission grid of the autoevolution and of the reconfiguration strategy $\mathcal{RS}_1()$ applied by LCS components. Moreover, in `RS_SAN` is also evaluated the function $\mathcal{RS}_2()$, by setting the extended places `P_SR2` and `F_SR2`. The evaluation of $\mathcal{AS}()$ is performed by the input function of the input gate `autoevDo`. The C++ procedure that returns the results of $\mathcal{AS}()$ is an user defined procedure based on the `GSL - GNU` Scientific Library[5]. The evaluation of $\mathcal{RS}_1()$ and $\mathcal{RS}_2()$ are performed by the input function of the input gate `enRS`. The C++ procedures that return the results of these functions are user defined procedures based on the library of `lp_solve`, a free linear (integer) programming solver[6].

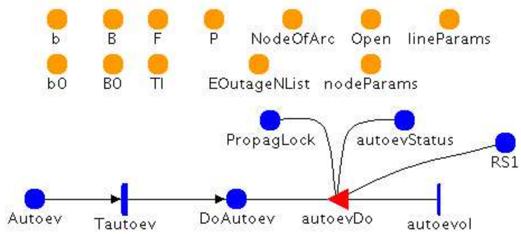

Fig. 8. The model `AUTOEV_SAN` for the autoevolution $\mathcal{AS}()$

Before to update the the values of **F**, the temperature of the line is also updated, based on the old value of **F** and on the instant of time of the previous update of the power through the line. The new configurations obtained by $\mathcal{AS}()$ or $\mathcal{RS}_1()$ (if needed) are applied after a small deterministic time.

The immediate activity `tInit` of the model `RS_SAN` is the only activity enabled in the overall model in the initial marking at the time 0 (when there

[5] http://www.gnu.org/software/gsl/

[6] http://lpsolve.sourceforge.net/5.5/

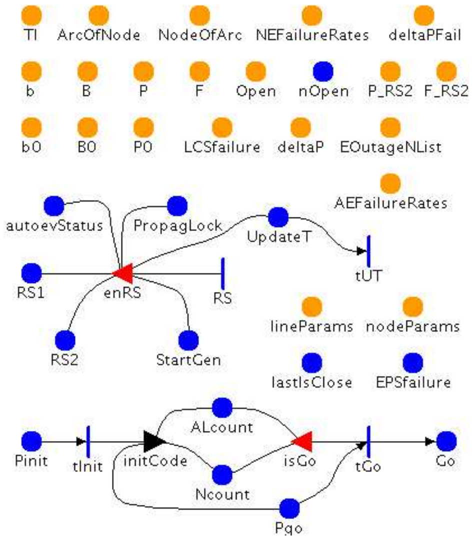

Fig. 9. The model RS_SAN for the reconfiguration strategies $\mathcal{RS}_1()$ and $\mathcal{RS}_2()$

is 1 token in the local place `Pinit`). The output gate `initCode` set the main parameters of the transmission grid, by executing the following C++ code:

```
const  char *configTfile="../../libMy/input/topologyConfig.txt";
const  char *configNfile="../../libMy/input/nodeParamsConfig.txt";
const  char *configLfile="../../libMy/input/lineParamsConfig.txt";
int i, j;

setupTopology(configTfile,  ArcOfNode, NodeOfArc, (unsigned short )
        nA, (unsigned short ) nN, b0, b, B0, B);
setupNode(configNfile, nodeParams, NEFailureRates, P0, P, (unsigned
        short ) nG, (unsigned short ) nS,
        (unsigned short ) nL, (unsigned short ) nN, varLoadIndex,
        alpha);
setupLine(configLfile, lineParams, AEFailureRates, (unsigned short ) nA);

setupP0F(scenarioPFtime0, lineParams, NodeOfArc, nodeParams, P0, b0,
        B0, P, F, (unsigned short ) nG, (unsigned short ) nS, (unsigned
        short ) nL, (unsigned short ) nN, LastActionTime);

setEquilibriumTemperatureOfLines(lineParams, Tl, 0, F, eta, nu, Tm); /* set the
    equilibrium temperature of each line at time 0 */

ALcount->Mark()=nA; /* start the setting of the indexes of the replicated lines */
Ncount->Mark()=nN; /* start the setting of the indexes of the replicated nodes */
Pgo->Mark()=1;
```

```
parameterCheck(nA, kAin, BigRate, AEFailureRates,
    deltaPGenPROpenTime, TTFGenDeltaP);
```

The main parameters of the considered system, including the description of the topology of the transmission grid, are described with three simple external files `topologyConfig.txt`, `nodeParamsConfig.txt` and `lineParamsConfig.txt`.

The atomic model RTS_SAN

`RTS_SAN`, shown in Figure 10, represents the states and the behavior of the Regional Tele-control System `RTS`, that is triggered to generate the new values for **P** and **F**. In particular, it models states where `RTS` works correctly, the states where omission failure occurs, the application of the reconfiguration strategy $\mathcal{RS}_2()$ and the list of the LCS that are failed during the evaluation of the reconfiguration strategy $\mathcal{RS}_2()$.

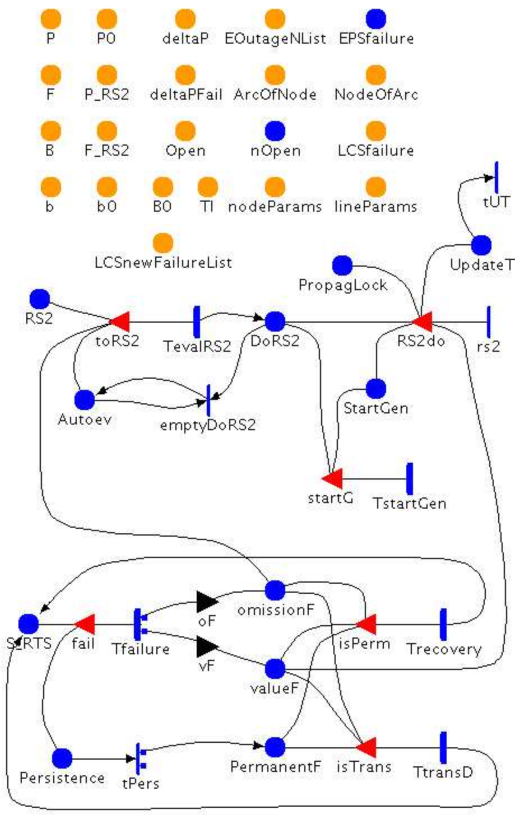

Fig. 10. The model `RTS_SAN` for the generic component `RTS`

When RTS is affected by an omission failure, the reconfiguration strategy $\mathcal{RS}_2()$ cannot be applied. Failures can be transient or permanent and are modeled like in `LCS_SAN`.

If during the evaluation of $\mathcal{RS}_2()$ new failures of `LCS` are occurred, then the new reconfiguration derived by $\mathcal{RS}_2()$ could require variations of the power associated to generators or loads, that cannot be modified. In this case, a new equilibrium state is needed to be obtained by triggering the autoevolution $\mathcal{AS}()$ based on the new no equilibrated state obtained by $\mathcal{RS}_2()$. Obviously, every time that in a model $\mathcal{RS}_2()$ is triggered, the list of the failed `LCS` has to be reset. When failures occur changing the status of `EI` during the evaluation of $\mathcal{RS}_2()$, then the evaluation of $\mathcal{RS}_2()$ is restarted based on the new topology generated by the new failure. Finally, like for `LCS_SAN`, the autoevolution $\mathcal{AS}()$ is triggered after a transient failure or a repair of `RTS`.

The atomic model COMNET_SAN

`COMNET_SAN`, shown in Figure 11, represents simplistically the states and the behavior of the public or private networks $TSOcomNet$ and $DSOcomNet$ of Figure 2. In particular, it models states where $TSOcomNet$ and $DSOcomNet$ work correctly (1 token in the place `S_CNET`), the states where (transient or permanent[7]) omission failure occur. The considered failures are modeled like in `LCS_SAN`.

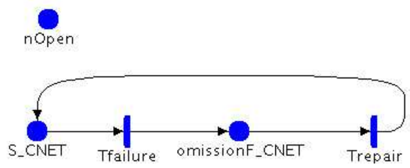

Fig. 11. The model `COMNET_SAN` for the generic component `COMNET`

The time to failures depends on the status of the transmission grid (the number of the open lines). The duration of a transient failure is exponentially distributed. The duration of the permanent failure is the deterministic time to repair the components $TSOcomNet$ and $DSOcomNet$, which depends on the status of the transmission grid (the number of the open lines).

3.3 Current status and next steps

Currently, the solution of the overall model of the addressed EPS instance is in progress using the tool Möbius, adopting some simple artificial scenarios, but well representative to show the impact of interdependencies on measures related

[7] To be completed.

with system blackouts. These results are expected to be highly useful to better understand the dynamics of the two interacting EI and ITCS infrastructures, and to help identifying appropriate architectural mechanisms to mitigate the impact of the revealed vulnerabilities.

We plan to exercise the developed framework on a wide variety of scenarios, including realistic control scenarios offered by the CRUTIAL consortium. This way, it will be possible to understand whether and where refinements/extensions of the proposed models are necessary, to cope with missing aspects or to make them more efficient from the solution point of view. The final goal is to pursue a truly general and composable modelling framework, adequate for the interdependencies evaluation in general EPS scenarios, and possibly extendible to interdependencies analysis in other critical infrastructure systems.

4 Conclusions

This paper has addressed the analysis of the electric power systems, among the major representatives of the critical infrastructures, with focus on the interdependencies that strongly characterize these systems. We have introduced a concrete approach to the definition of an evaluation framework tailored to electric power system, which is currently under development in the context of the European project CRUTIAL. Specifically, the identification of the logical components composing the EPS infrastructures as well as their interdependencies have been outlined, mainly reporting from a previous work by the authors. On the basis of the characteristics of the involved components and of the challenges in evaluating CI, major features to be possessed by a model-based evaluation framework for quantitative analysis of the interdependencies and their impact on the correct operation of EPS systems have been pointed out. Then, a specific instance of the EPS system has been more concretely tackled, by providing i) the abstract view of the models representing the several involved components and the mechanisms through which they interact, and ii) their concrete implementation using Stochastic Activity Networks formalism. Actually, the developed models are reusable template models, which can be assembled to represent a number of specific scenarios in the EPS sectors (theoretically, the proposed building block models should allow representing all the possible EPS configurations involving (a subset of) the addressed components). An evaluation campaign is currently undertaken to derive quantitative assessment of the impact of interdependencies on blackouts related indicators. The results are expected to be very useful, first to guide the possible refinement/extension of the proposed modelling framework, so as to have a solid base for a better understanding of EPS vulnerabilities, thus leading to enhanced design choices for EPS protection at architectural level. Comparison with the results obtained through the EPS simulator [10], under development by the research group the authors belong to, will be a valuable source for cross validation of the two assessment methods.

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